

Large-Eddy and Flight Simulations of a Clear-Air Turbulence Event over Tokyo on 16 December 2014

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ABSTRACT: In this study, a clear-air turbulence event was reproduced using a high-resolution (250 m) large-eddy simulation in the Weather Research and Forecasting (WRF) Model, and the resulting wind field was used in a flight simulation to estimate the vertical acceleration changes experienced by an aircraft. Conditions were simulated for 16 December 2014 when many intense turbulence encounters (and one accident) associated with an extratropical cyclone were reported over the Tokyo area. Based on observations and the WRF simulation, the turbulence was attributed to shear-layer instability near the jet stream axis. Simulation results confirmed the existence of the instability, which led to horizontal vortices with an amplitude of vertical velocity from +20 to -12 m s^{-1} . The maximum eddy dissipation rate was estimated to be over 0.7, which suggested that the model reproduced turbulence conditions likely to cause strong shaking in large-size aircraft. A flight simulator based on aircraft equations of motion estimated vertical acceleration changes of +1.57 to +0.08 G on a Boeing 777-class aircraft. Although the simulated amplitudes of the vertical acceleration changes were smaller than those reported in the accident (+1.8 to -0.88 G), the model successfully reproduced aircraft motion using a combination of atmospheric and flight simulations.

KEYWORDS: Turbulence; Kelvin–Helmholtz instabilities; Large-eddy simulations; Transportation meteorology

1. Introduction

Atmospheric turbulence is a serious threat to aviation safety, and turbulence-related events have caused many aviation accidents. In particular, clear-air turbulence (CAT) is generated outside convective clouds, making it difficult for onboard radars to detect. Kelvin–Helmholtz (KH) instability can explain most causes of high-altitude ($\sim 10 \text{ km}$) CAT (Ellrod and Knapp 1992). Turbulent eddies from KH instability decrease in size due to energy cascades, eventually reaching aircraft-size, and it is these small eddies that are mainly responsible for aircraft turbulence (Kim and Chun 2010; Joseph et al. 2004; Sekioka 1970). Breaking of gravity waves propagating from terrain or convection can also be a source of CAT (Sharman et al. 2012; Kim and Chun 2010; Trier et al. 2012; Lane et al. 2003; Lane and Sharman 2008). Other potential sources of turbulence include shear instability or shallow convective instabilities arising from modifications to environmental vertical shear and static stability in mesoscale outflows originating from horizontally distant convections (Trier and Sharman 2016; Trier et al. 2020).

Several organizations have developed technologies to prevent CAT-related accidents. For example, the Japan Aerospace Exploration Agency (JAXA) has developed and tested aircraft-mounted light detection and ranging (lidar) systems that can detect CAT ahead of aircraft and a framework to mitigate the impact of CAT using existing aircraft control surfaces (ailerons, elevators, etc.) in conjunction with the lidar

observation information (Inokuchi et al. 2009, 2016; Machida et al. 2018; Kikuchi et al. 2020; Inokuchi and Akiyama 2019). The JAXA lidar system observes an infrared laser reflected from aerosol particles in front of an aircraft and estimates their speed, which is assumed to be the local wind speed, on the basis of the Doppler shift. Flight tests conducted in 2017 (Matayoshi et al. 2018) demonstrated the capability of the lidar system to observe CAT that is over 20 km away below 5000-ft and over 10 km away at 40000-ft altitudes ($1000 \text{ ft} \approx 305 \text{ m}$). The German Aerospace Center [Deutsches Zentrum für Luft- und Raumfahrt (DLR)] has also developed an aircraft-mounted lidar system with an improved algorithm to reproduce the forward wind field from lidar observations and validated their devices and attitude control systems using numerous experiments, including flight tests (Schmitt et al. 2007; Veerman et al. 2014; Fezans et al. 2019; Vrancken et al. 2016).

One challenge with the aforementioned developments is that not much actual measurement data of realistic CAT can be obtained. While flight testing is an important process to determine whether lidar devices can meet the performance specifics required under real flight conditions, the amount of data collected during flight tests can be insufficient because the number of flight tests is limited or because the testing aircraft rarely encounter strong turbulence. For example, there were few encounters with intense CAT during 33 h of flight tests conducted by DLR in 2013 (Veerman et al. 2014), and they succeeded in validating their system based on encounter data of less intense CAT. Validation of lidar systems in an environment with intense CAT, which is one of the major causes of aviation accidents, is generally difficult.

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By establishing a method to simulate CAT and estimate its impact on aircraft, virtual flight testing under near realistic conditions becomes possible, facilitating more efficient development. Reproduction of detailed CAT structures using numerical methods enables virtual flight testing by inputting simulated CAT into aircraft equations of motion (Takase et al. 2017; Kato et al. 1982; McRuer et al. 1973) that have six (three in translation and three in rotation) degrees of freedom, and solving them. For a limited number of future flight tests, a preliminary investigation may be possible in numerical space to analyze the aircraft and attitude control system response when encountering CAT, and to validate the algorithm for wind field estimation from lidar observations. To obtain a CAT simulation close to reality, a numerical weather prediction model can be used to reproduce and forecast realistic weather.

The purpose of this study is to investigate the feasibility of virtual flight testing by performing a flight simulator in a wind field with CAT obtained from detailed simulations of turbulence using the large-eddy simulation (LES) option in the Weather Research and Forecasting (WRF) Model (Skamarock et al. 2008). Virtual flight testing requires a three-dimensional and high-resolution wind field; however, these types of data are not commonly available. Although there are several examples of numerical simulations of turbulence, including CAT (Trier et al. 2020; Trier and Sharman 2016; Trier et al. 2012; Sharman et al. 2012; Kim and Chun 2010), these efforts resolve turbulence using a combination of a planetary boundary layer and horizontal mixing parameterizations. Most LES applications for numerical weather prediction have focused on turbulence within the boundary layer (Wurps et al. 2020; Onishi et al. 2019; Shamsoddin and Porté-Agel 2017; Moeng et al. 2007), and there are almost no examples of LES at high altitudes. Misaka et al. (2008) succeeded in reconstructing a two-dimensional wind field using LES and variational data assimilation of flight data from past CAT encounters. However, their reconstruction was based on maximum likelihood estimation and not on physical equations, and it was not confirmed whether the wind field was meteorologically consistent. Moreover, there are no examples of flight simulations in a wind field simulated by a numerical weather prediction model.

To create a detailed turbulence dataset for input into a flight simulator, we used WRF to reproduce the conditions that led to an accident involving a Boeing 777-200 aircraft that occurred over Tokyo at 1035 UTC 16 December 2014 (1935 LST 16 December 2014), hereinafter referred to as the “target case” as shown in Fig. 1. The target case was chosen because the aircraft experienced significantly large acceleration changes (from +1.8 to -0.88 G; Japan Transport Safety Board 2016), there were many pilot reports (PIREPs) in the vicinity that detailed strong shaking from 730 to 1100 UTC, and it was relatively easy to acquire observation data given the dense observations around Tokyo. As shown in Fig. 1, area A includes the location of the target case and is the location of a particularly large number of PIREPs. We focus on area A in section 2.

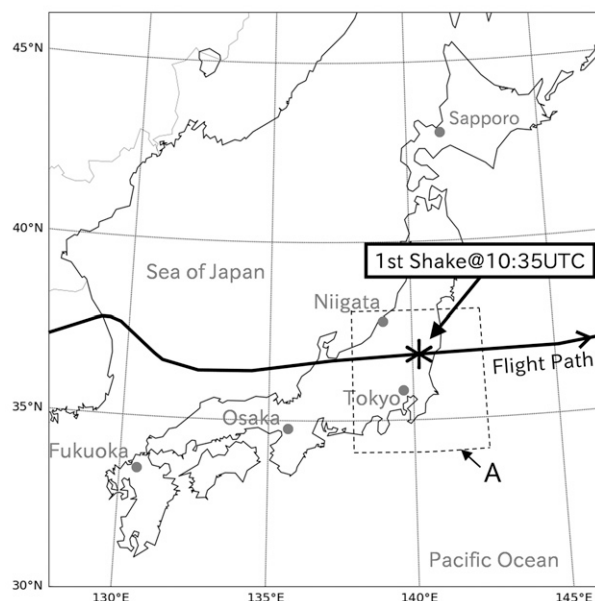


FIG. 1. Overview of the accident involving a Boeing 777-200 aircraft that occurred over Tokyo at 1035 UTC 16 Dec 2014 (1935 LST 16 Dec 2014; the target case). The flight path, time, and location of the target case are from the accident investigation report (Japan Transport Safety Board 2016).

2. Turbulence observations

On 16 December 2014, a developing extratropical cyclone moved northeastward along the southern coast of Japan. Most turbulence encounters were reported from 0730 to 1100 UTC in area A on Fig. 1 at the time when the cyclone was approaching Tokyo (Fig. 2; 1200 UTC 16 December 2014). Strong southwesterly winds dominated at an altitude of 300 hPa (Fig. 7a, described in more detail below).

We refer to the Japan Airlines (JAL) PIREP to discuss the reports of turbulence encounters. In a PIREP, the intensity of turbulence is judged by the captain on the basis of the cabin condition, experience, and controllability. The JAL PIREP data contain six levels of intensity, as described in Table 1 (Japan Civil Aviation Bureau 2009). In this study, we used moderate-or-greater (MOG) turbulence reports to investigate the general spatiotemporal trends in the PIREP data.

Figure 3 shows that the location of the MOG PIREPs gradually moves northeastward over time, in the same direction as the cyclone. Most of the PIREPs were located between an altitude of 7 and 9 km (Fig. 3d), which approximately corresponds to the orange-colored points (FL251–FL300, or approximately 8–9 km altitude) in Figs. 3a–c. The target case (8.2 km altitude; 1035 UTC) was also located in this altitude range.

Figure 4 shows radiosonde observation data from 1200 UTC 16 December 2014 at Tateno (Fig. 3c), the closest radiosonde location to the targeted case. A strong vertical wind shear is apparent at 7–9 km, just under the jet stream axis located at an altitude of 10 km. The range of 7–9 km corresponds to the altitude at which a large number of MOG-level PIREPs were

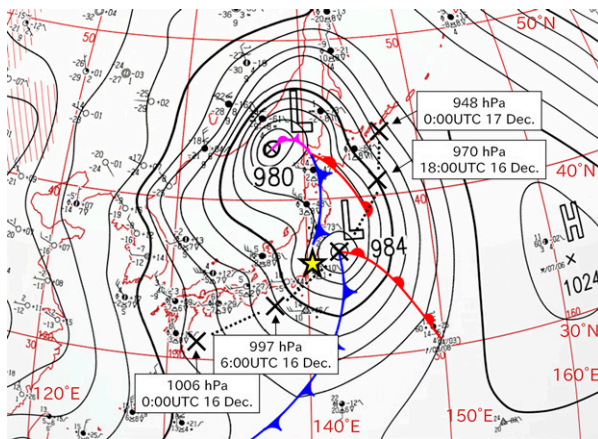


FIG. 2. Surface weather map at 1200 UTC 16 Dec 2014 and the path of the low pressure system over Japan (from the JMA). The star denotes the location of the target case.

reported. Therefore, we suggest that the vertical wind shear below the jet stream axis contributed to turbulence generation at this altitude.

Figure 5 shows the Japan Meteorological Agency (JMA) radar echo-top height at 0900 (Fig. 5a), 1030 (Fig. 5b), and 1200 (Fig. 5c) UTC 16 December 2014, corresponding to the PIREPs shown in Fig. 3. The precipitation area moves northeastward along with the cyclone, and the echo-top height is below 8 km in most areas of area A. Figure 5b shows that the altitude of the target case (8.2 km) is approximately the same as the observed echo-top height. In the accident report by the Japan Transport Safety Board (2016), the crew described the conditions at the time of the encounter as “It was still dark and thin clouds were sometimes seen outside the windows of the cockpit and an airborne radar did not display the clouds expected to cause the big shake” (p. 3), suggesting that the aircraft was flying outside the precipitation cloud at the time.

To summarize, cyclone-related MOG-level turbulence events were frequently reported at 7–9 km in the vicinity of Tokyo (Fig. 1; area A), and the target case occurred at the same time. Radar and radiosonde observation data suggest that the 7–9-km altitude was between the cloud top (Fig. 5)

and the jet stream axis (Fig. 4). The captain’s testimony suggests that the flight was outside the precipitation clouds. Therefore, some local turbulence existed above (not inside) precipitation clouds in area A, and the target case occurred when the aircraft was flying around the cloud top.

3. Model configuration

We conducted numerical experiments with horizontal grid spacings of 3, 1, and 0.25 km using the WRF-ARW, version 3.8, model. Figure 6 shows the computational domain and terrain height. JMA Mesoscale Model (MSM) analysis (5-km grid spacing; JMA 2013) defined the initial/lateral boundary conditions above the ground. NCEP $1^\circ \times 1^\circ$ reanalysis (NOAA/NCEP 2000) was used for the soil conditions. The model incorporates 100 vertical layers from the surface to 100 hPa; vertical grid spacing increases with altitude and becomes constant ($dz \sim 180$ m) above approximately 2 km. The start time of the numerical experiments was 1800 UTC 15 December 2014, approximately 16.5 h prior to the onset of the turbulence events. We turned on the two-way nesting option (feedback = 1).

The WRF single-moment 6-class graupel scheme (Hong and Lim 2006) was used for microphysics parameterization. This scheme predicts the mixing ratios of water vapor, cloud water, rainwater, ice, snow, and graupel. A cumulus parameterization scheme was not used in the present study. The MYJ turbulent kinetic energy (TKE; Janjić 1994, 2001), horizontal Smagorinsky, and Monin–Obukhov schemes governed subgrid vertical mixing, subgrid horizontal mixing, and surface-layer physics, respectively. The MYJ TKE scheme solves the one-dimensional equation for subgrid TKE and diagnostic equations for the vertical subgrid-scale fluxes not only in the planetary boundary layer but also in the free atmosphere.

For domain 3, an LES was performed using a Smagorinsky first-order closure scheme (Moeng et al. 2007; Mirocha et al. 2010). Subgrid-scale mixing in the horizontal and vertical directions were predicted by subgrid-scale stress, which is the product of wind shear and eddy viscosity. The eddy viscosity depends on the local grid size and the magnitude of the wind shear. For horizontal grid interval, $dx = 250$ m can resolve the first vortices that appear from the KH instability, the largest eddies that affect the motion of aircraft, with several points in this case.

4. Results

a. Synoptic-scale simulation (domain 1)

Figure 7 compares simulated 300-hPa wind and height and radar echo-top height with JMA MSM data and radar observations, respectively, for domain 1 at 1000 UTC (Fig. 7a,b) and 1030 UTC (Fig. 7c,d) 16 December 2014. The WRF simulation estimates the MSM analysis well. The observed echo-top height was reproduced well in domain 1 for its maximum height (8–10 km) around 139° – 141° E, 35° – 37° N.

Figure 8 shows the wind field at $z = 8$ km at 1000 UTC in domain 1, representing an altitude close to that of the target case. Small-scale up/downdrafts are apparent over an area of high total cloud condensate and form wave trains. Up/

TABLE 1. Turbulence intensity categories defined by JAL and the Federal Aviation Administration (FAA).

FAA intensity	JAL intensity	Impact on comfort
Smooth	Smooth	No impact; feels like being on the ground
Light	Light minus, light, and light plus	Slight vibration but still possible to walk or stand in the cabin
Moderate	Moderate	Difficult to walk or stand; objects move around in the cabin
Severe	Severe	Impossible to walk or stand
Extreme		(Risk of structural failure on aircraft)

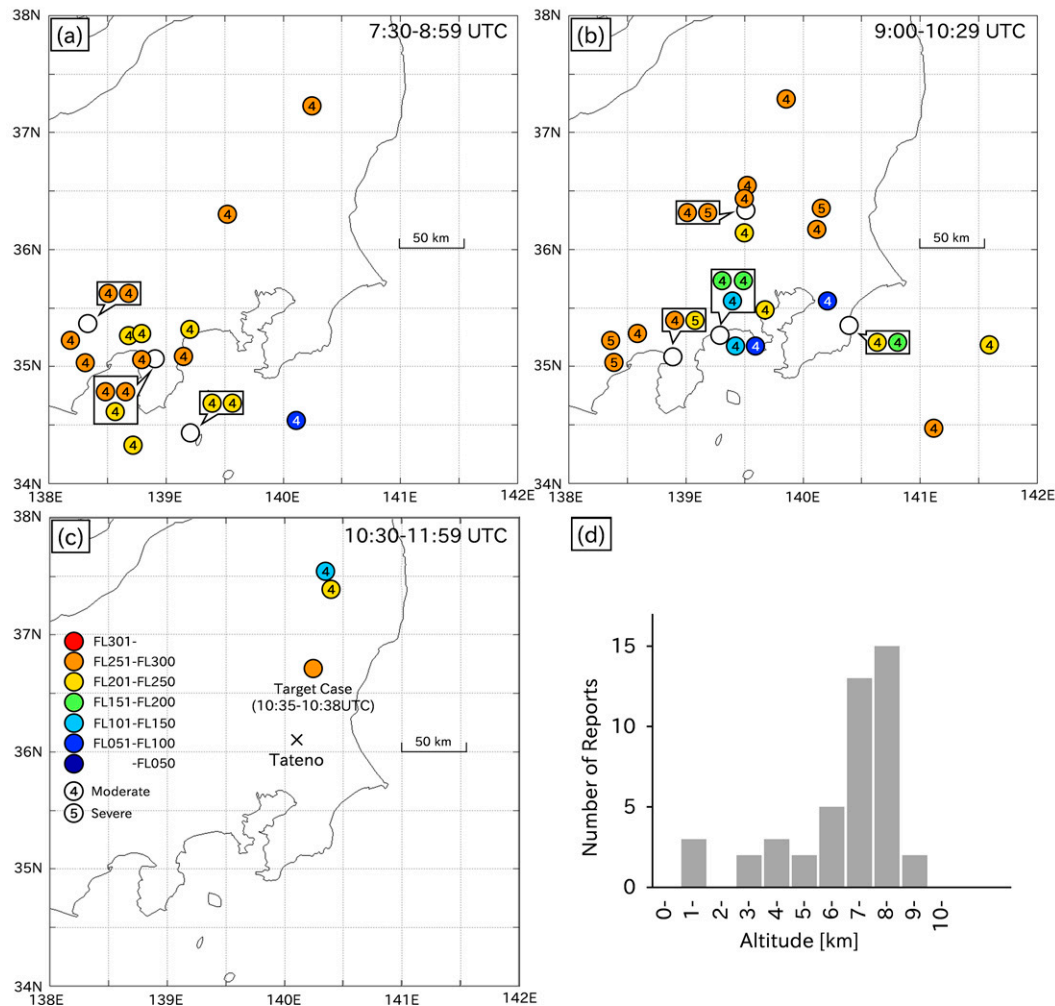


FIG. 3. MOG-turbulence-level pilot reports in area A (a) from 0730 to 0859 UTC, (b) from 0900 to 1029 UTC, and (c) from 1030 to 1159 UTC, along with (d) a histogram of report altitudes; (c) includes the location of the target case and the radiosonde flight (Tateno, denoted by an X). The turbulence intensity and altitude [flight level (FL)] are represented by numbers (moderate = 4 and severe = 5) and colors, respectively. FL is altitude information at 100-ft (≈ 30.5 m) intervals, which is represented by a three-digit number.

downdrafts possibly due to vertically propagating mountain waves from the mountainous region along the 138.5°E line (near the west edge in Fig. 8) are also apparent; however, these waves have a different spatial scale than the wave trains. Therefore, over the region where many MOG-level PIREPs were reported, regular fluctuations in vertical wind speed, not attributed to terrain, are widely distributed over the clouds.

b. Turbulence generation mechanism (domain 1)

To understand the generation of wave trains, Fig. 9 shows the cross section A–B along one of the wave trains observed in Fig. 8. The cross section confirms a band with Richardson numbers (Ri) < 0.25 , (green shading) at 6–8 km altitude, and the fluctuation of vertical winds (red/blue shading) centered around the band extends to 12–13 km through gravity waves. Figure 9a highlights A' as the height region where most of the

PIREPs were reported, and it is apparent that A' corresponds to vertical wind fluctuations along the $Ri < 0.25$ band. Ri is a dimensionless number defined by

$$Ri = \frac{N^2}{VWS^2}, \quad (1)$$

where N^2 is the Brunt–Väisälä frequency and VWS represents the vertical wind shear. For two-dimensional stratified shear flow, $Ri < 0.25$ is a necessary condition of shear-layer instability (Nappo 2002; Drazin and Reid 1981).

According to Scorer (1969), the range of wavelengths that gives a positive growth rate to a sinusoidal wave on a vertical shear layer of Δz width can be approximated using Ri , as shown in

$$\lambda < \lambda_{\max} = \frac{\pi \Delta z}{Ri}. \quad (2)$$

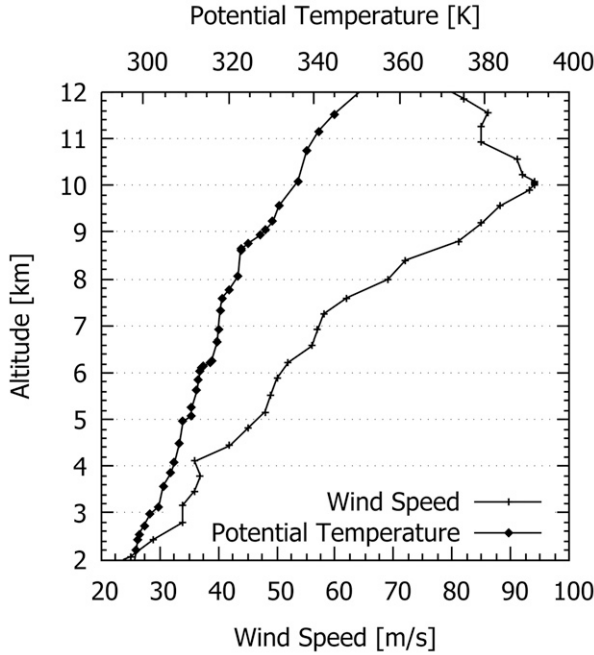


FIG. 4. Radiosonde data from Tateno (refer to Fig. 3c) at 1200 UTC 16 Dec 2014.

In Fig. 9a, the average wavelength of waves at $x = 25\text{--}70$ km, where the direction of the wave train is parallel to the cross section, is $\lambda = 14.7$ km. The average width of the $Ri < 0.25$ region (Δz) is 1.26 km. The $\lambda_{\max} = 15.9$ km obtained based on these parameters suggests that the wave trains were generated by shear-layer instability. Because the $Ri < 0.25$ region is located near the upper end of the region where the potential temperature gradient becomes small (Fig. 9a), and in the large vertical shear region (Fig. 9b), we suggest that both the reduction of atmospheric stability and the reinforcement of vertical wind shear would contribute to the reduction of the Ri .

Figure 10 shows shear vectors from 6 to 8 km, calculated by subtracting 6-km wind vectors from 8-km wind vectors. Because the 6-km wind vector is approximately 35 m s^{-1} northwesterly and the 8-km vector is approximately 63 m s^{-1} west-southwesterly over Tokyo (35.75°N , 139.75°E), the shear vector is westerly. Over Tokyo, the shear vector and wave train are almost parallel. Figure 11 shows simulated radar reflectivity and potential temperature along the vertical cross section A–B (Fig. 8). We can see that the wave train along the $Ri < 0.25$ band is located above the high radar reflectivity region. This means that the $Ri < 0.25$ region, the possible source of the turbulence, is placed over the precipitation clouds, consistent with the captain's testimony in section 2.

c. LES results (domain 3)

Figure 12 shows the horizontal distribution of vertical wind speeds at 8.2 km, the altitude of the target case, and 2-km radar reflectivity at 1021 UTC 16 December 2014 for the domain-3 simulation. Similar to Fig. 8, wave trains are apparent over regions of high total cloud condensate. Figure 13

shows the simulated Richardson number, potential temperature, and vertical winds along the vertical cross section C–D (Fig. 12). The magnitude of the vertical wind fluctuation peaks at an altitude of approximately 6–8 km, where the $Ri < 0.25$. It is apparent that the strong vertical wind fluctuation reaches an altitude of 8.2 km, the flight level of the target case. In terms of magnitude, the domain-1 simulation estimates a small fluctuation of $\pm 8 \text{ m s}^{-1}$ or less. In contrast, the domain-3 simulation reproduces fine-scale fluctuations possibly formed by breaking of the wave trains and estimates an amplitude from $+20$ to -12 m s^{-1} (refer to section 5). Furthermore, domains 1 and 3 estimated different values for the spatial frequency of the waves. As mentioned in section 4b, domain 1 estimated a wavelength of approximately 15 km for the first instability mode from shear-layer instability. However, in domain 3 the wavelength of waves thought to be the first instability mode is approximately 10 km, 30–40 times the horizontal grid size. Overturning isentropes are found at an altitude of approximately 6–8 km, indicating wave breaking and the likely onset of turbulence.

We investigate the intensity of turbulence simulated in domain 3 by computing the energy dissipation rate (EDR) ε to the power of $1/3$; $\varepsilon^{1/3}$ is known as an objective index of turbulence intensity (International Civil Aviation Organization 2016). According to Schumann (1991), ε can be approximated using subgrid TKE, as shown in

$$\varepsilon \approx \frac{0.84e^{3/2}}{\Delta}, \quad (3)$$

where Δ is the length scale based on grid spacing $\Delta = (\Delta x \Delta y \Delta z)^{1/3} = 224 \text{ m}$, and we use resolved TKE as e to make a rough estimation of EDR. The resolved TKE is calculated for domain 3 based on locally averaged wind components using a Gaussian filter [Eqs. (4)–(6)] subtracted from local wind components. We use a Gaussian filter to obtain the local average that follows the wind speed variation well and removes high-frequency fluctuations. Here, σ is the radius of localization, which is set to 500 m to evaluate TKE due to near aircraft-size wind variation:

$$e(\text{lon}, \text{lat}) = \frac{\sum_{i=1}^3 [u'_i(\text{lon}, \text{lat})]^2}{2}, \quad (4)$$

$$u'_i(\text{lon}, \text{lat}) = u_i(\text{lon}, \text{lat})$$

$$- \sum_{x,y} F(x,y) u_i(\text{lon} - x, \text{lat} - y), \quad \text{and} \quad (5)$$

$$F(x,y) = \frac{1}{(\sqrt{2\pi}\sigma)^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \quad (6)$$

Figure 14 shows the horizontal distribution of $\varepsilon^{1/3}$ and the $\pm 1 \text{ m s}^{-1}$ isoline of the vertical wind speed at 1021 UTC 16 December 2014 in the same area, as shown in Fig. 12. EDR is large along with the wind variation due to the shear-layer instability, which peaks in the region of large wind fluctuations (see Figs. 12 and 14). In most areas, $\varepsilon^{1/3}$ is less than $0.7 \text{ m}^{2/3} \text{ s}^{-1}$, which is the threshold value between moderate and severe turbulence categories described in the International Civil Aviation

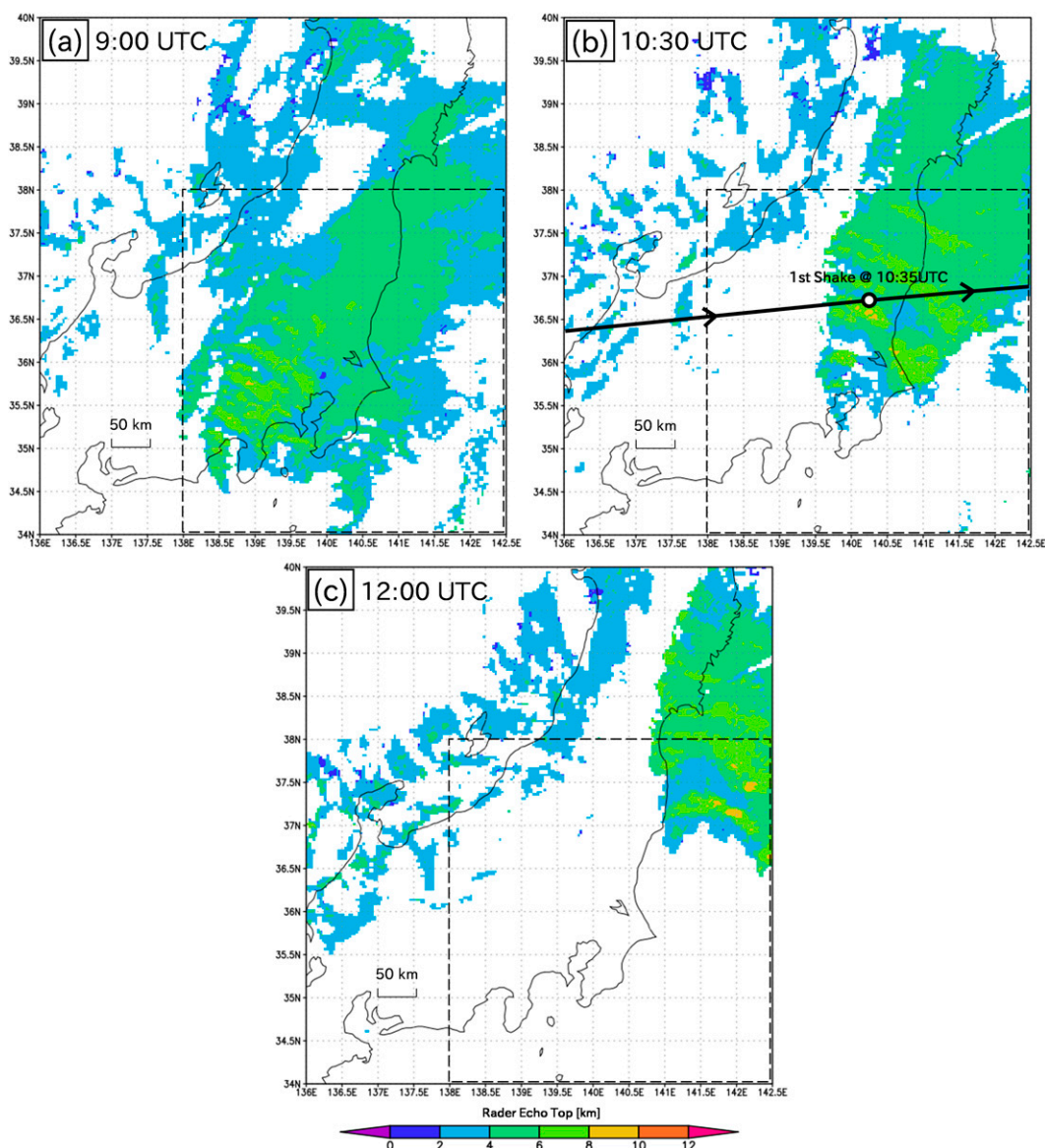


FIG. 5. Observed echo-top height at (a) 0900, (b) 1030, and (c) 1200 UTC 16 Dec 2014. In (b), the flight path and the turbulence location of the target case are superimposed on the observed echo-top height. The outline of area A is shown as a dotted rectangle.

Organization (ICAO) Annex 3 (International Civil Aviation Organization 2016). However, some values exceed 0.7, and the maximum value is 0.86 in the white dashed circle. For computation with $\sigma = 1000$ m, which is sufficiently larger than aircraft size, the maximum EDR increases to 1.49, because the local wind fluctuation becomes larger.

5. Simulated aircraft response

a. Methods

We use the aircraft equations of motion (Kato et al. 1982; Takase et al. 2017; McRuer et al. 1973) to estimate the vertical

acceleration experienced by an aircraft flying in an airspace with the turbulence reproduced in section 4c. Aircraft motion can be approximated by solving the six-degrees-of-freedom equations of motion about an aircraft's center of gravity, as shown in Eqs. (7)–(12). Equations (7)–(9) solve the motion by translational forces (F_x , F_y , and F_z), and Eqs. (10)–(12) solve the rotational motion by moments (M_x , M_y , and M_z). The equations are defined on a coordinate system whose origin is located at the aircraft's center of gravity. The variables m and I represent the aircraft mass and moment of inertia, respectively. These forces and moments are caused by gravity or the interaction between the aircraft and air. Figure 15a defines the coordinate system and directions of the forces and moments:

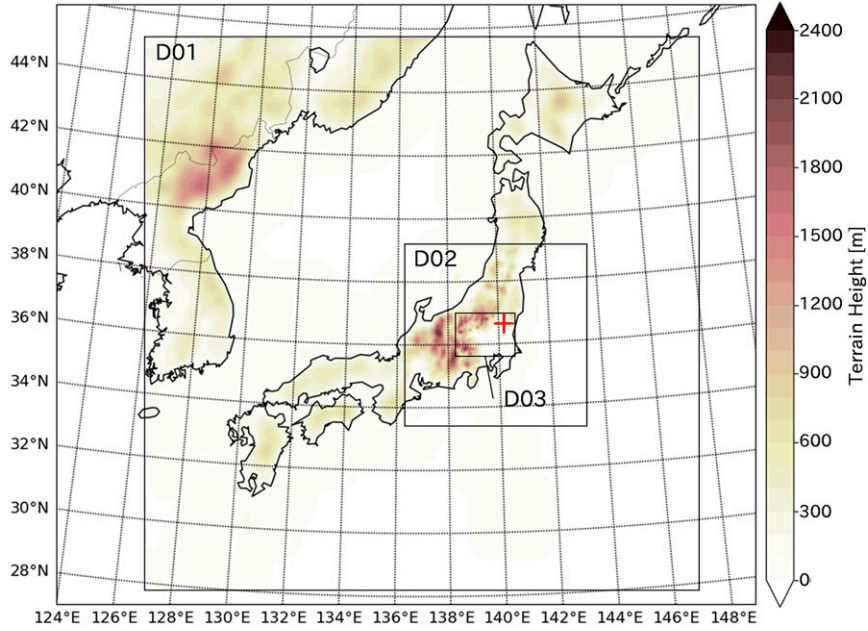


FIG. 6. Model domains 1 (1920×1920 km), 2 (649×649 km), and 3 (195×150 km). Domain 3 resolves part of area A, which includes the location of the target case (red cross).

$$m \frac{du}{dt} = F_x - mg \sin \theta, \quad (7)$$

$$m \frac{dv}{dt} = F_y, \quad (8)$$

$$m \frac{dw}{dt} = F_z, \quad (9)$$

$$I_x \frac{dp}{dt} - I_{xz} \frac{dr}{dt} = M_x, \quad (10)$$

$$I_y \frac{dq}{dt} = M_y, \quad \text{and} \quad (11)$$

$$I_z \frac{dr}{dt} - I_{xz} \frac{dp}{dt} = M_z. \quad (12)$$

In this study, we only partially solved these equations to simulate vertical acceleration changes. Equations (7)–(12) can be separated into two independent groups of equations by linearization of the external force terms using the methods introduced in the [appendix](#): equations for longitudinal motions that include vertical acceleration information and equations for lateral motions. The former group contains the following three equations:

- 1) Eq. (7), the temporal change in the horizontal speed u of the aircraft due to thrust, drag, and the x -axis component of the gravity force,
- 2) Eq. (9), the temporal change in the vertical speed w of the aircraft due to the lifting force, and
- 3) Eq. (11), the temporal change of the rotational angular velocity about the y axis due to the aerodynamically generated angular moment about the y axis.

We do not solve Eq. (7) because changes in the angle of attack (the degree between the x axis and the aircraft direction of travel α ; refer to [Fig. 15b](#)) caused by vertical wind changes generally have a much greater impact on vertical acceleration variation relative to horizontal wind changes. This means that the horizontal speed of an aircraft is set to a constant value of U_0 . The equations of motion used in this study are given by Eqs. (13) and (14):

$$\frac{d}{dt} \begin{bmatrix} \alpha \\ q \end{bmatrix} = \begin{bmatrix} \frac{Z_\alpha}{U_0} & 1 + \frac{Z_q}{U_0} \\ \frac{M'_\alpha}{U_0} & \frac{M'_q}{U_0} \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \frac{1}{U_0} \begin{bmatrix} Z_u & \frac{Z_\alpha}{U_0} - Z_q \frac{\partial}{\partial X} \\ \frac{M'_u}{U_0} & \frac{M'_\alpha}{U_0} - M'_q \frac{\partial}{\partial X} \end{bmatrix} \begin{bmatrix} u_g \\ w_g \end{bmatrix} \quad \text{and} \quad (13)$$

$$M'_u = \left(M_u + \frac{M_{\alpha'} Z_u}{U_0} \right), \quad M'_\alpha = \left(M_\alpha + \frac{M_{\alpha'} Z_\alpha}{U_0} \right),$$

$$M'_q = \left[M_q + \frac{M_{\alpha'} (U_0 + Z_q)}{U_0} \right]. \quad (14)$$

The left-hand side in Eq. (13) indicates the temporal changes, the first term on the right-hand side is the aerodynamic force depending on the attitude of the aircraft (α and q), and the second term is the additional force due to turbulence (u_g and w_g). [Figure 15b](#) shows the variables in the equations. The angle of attack α is approximated from the vertical speed of aircraft w by using $\alpha = \text{atan}(w/U_0) \approx w/U_0$, under the assumption of a very small α . The q is the angular velocity along the y axis ($q = d\theta/dt$). Both α and θ have the same unit (rad); however, the

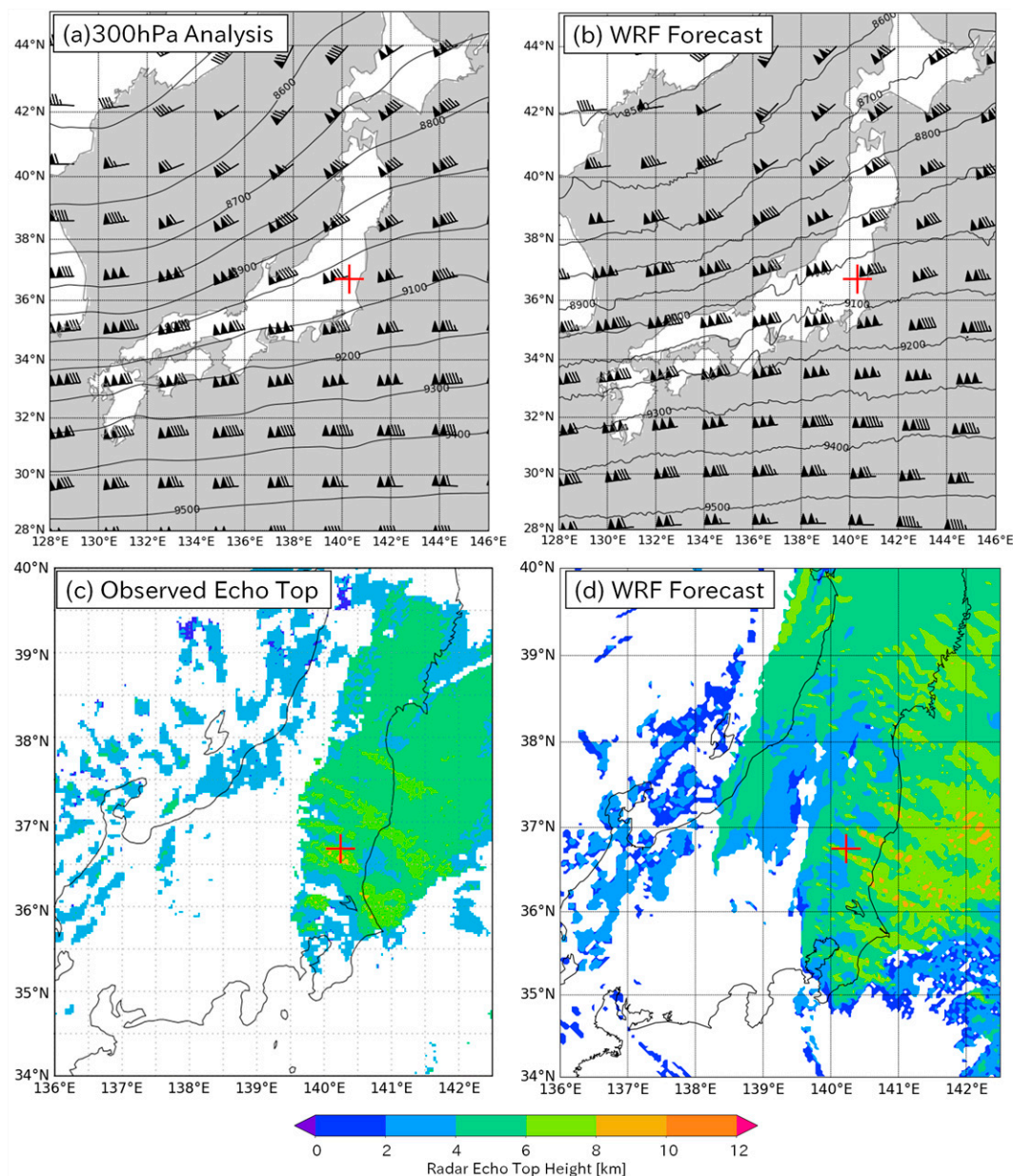


FIG. 7. Comparisons of the (a) JMA MSM analysis and (b) WRF domain-1 simulated 300-hPa wind and height at 1000 UTC 16 Dec 2014 and (c) observed and (d) WRF domain-1 simulated radar echo-top height at 1030 UTC 16 Dec 2014. The location of the target case is denoted by a red cross. Radar reflectivity tends to be weaker in the distant sea.

former is the angle based on the aircraft's direction of travel, whereas the latter is the angle based on the Earth coordinate system. In other words, α defines the aircraft's direction of travel, and θ defines the angle of the aircraft from Earth's coordinate system. The coefficients (such as Z_α and M_q) for expressing aerodynamic forces and moments in combination with these state variables (α and $q = d\theta/dt$) or turbulence (u_g and w_g) are unique to each aircraft. The definitions of the coefficients, estimation, and validity of the estimation method are given in the [appendix](#).

b. Computational conditions

We set the conditions for the flight simulation to be close to those of the target case. We estimated the aerodynamics of the Boeing 777-200 (coefficients Z_α , M_q , etc.) and substituted them into Eqs. (13) and (14). The aircraft flies at $U_0 = 258 \text{ m s}^{-1}$ along the path from I to J that crosses the region of peak EDR (the white dash circle in [Fig. 14](#)), on the 8.2-km cross section at 1021 UTC 16 December 2014. [Figure 16](#) shows the initial location of the assumed aircraft and its flight direction.

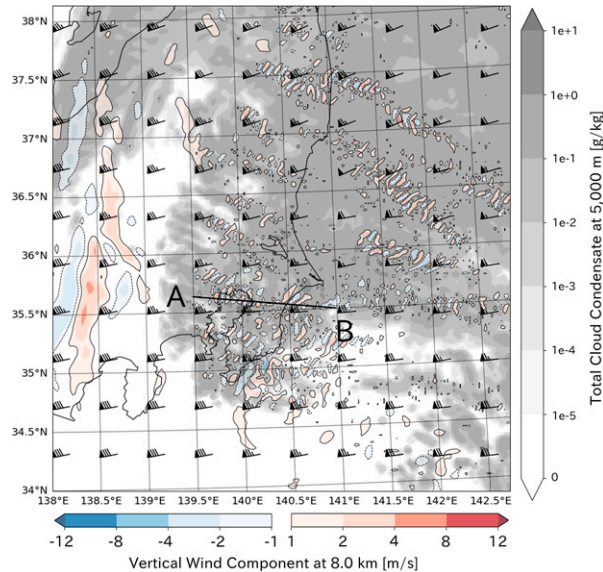


FIG. 8. WRF domain-1 simulated horizontal wind barbs (long barbs 10 m s^{-1} , short barbs 5 m s^{-1}) and $|\text{vertical wind component}| > 1 \text{ m s}^{-1}$ (red/blue shading) at an altitude of 8.0 km (the approximate altitude of the target case), and 5-km total cloud condensate (gray shading) at 1000 UTC 16 Dec 2014. The transect AB refers to the location of the vertical cross sections presented in Fig. 9, below.

We solve Eqs. (13) and (14) for 70 s (18.3 km), from the initial point *I* in the atmospheric field with the time fixed at 1021 UTC 16 December 2014, using the WRF simulation results as turbulence (u_g , w_g). In addition, the aircraft does not have any control rules; we do not assume the pitching control embedded in the Boeing 777 aircraft (C^* law; Spitzer 2001) in this study.

c. Results

Figure 17a shows the simulated aircraft response. While flying a distance of 18.3 km, the aircraft experiences up to $|\Delta G| = 1.49 \text{ G}$ due to a large wind change at 12–14 km. The maximum and minimum accelerations are 1.57 and 0.08 G, respectively, and the maximum deviation from 1 G is 0.92 G, which is classified as moderate turbulence according to Aircraft Meteorological Data Relay (AMDAR) turbulence categories (WMO 2003). This suggests that WRF-LES data and flight simulations can reproduce a situation that causes moderate shaking in the AMDAR scale on a Boeing 777-class aircraft without pitching control.

Figure 17b shows the histories of the observed vertical/horizontal winds and the vertical acceleration experienced by the aircraft. The aircraft experiences a local minimum acceleration of 0.08 G due to vertical wind changes from $+20$ to -12 m s^{-1} around $t = 50 \text{ s}$. The contribution of each term in the equations of motion to the acceleration changes is also shown. Acceleration due to horizontal wind changes $G_{ug} = (Z_u/U_0)u_g$ (red dashed line) is very small relative to acceleration due to vertical wind changes $G_{wg} = (Z_w/U_0 - Z_q \partial/\partial X)/U_0 w_g$ (blue dashed line)

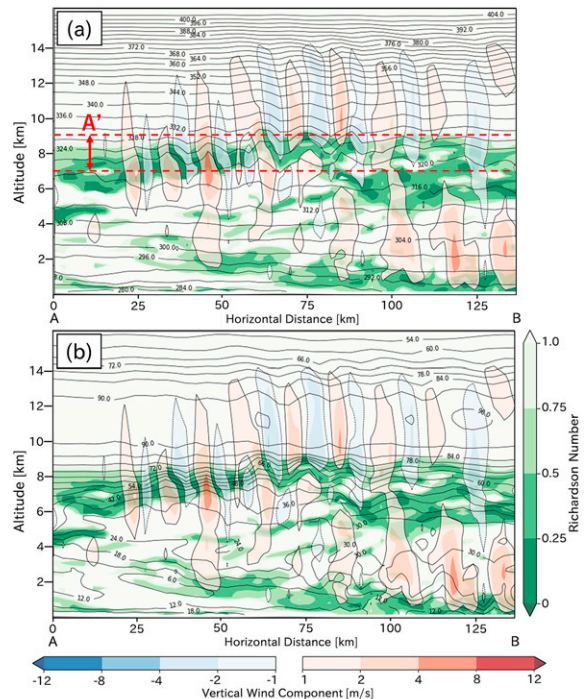


FIG. 9. WRF domain-1 simulated Richardson number (Ri ; green shading) and the $|\text{vertical wind component}| > 1 \text{ m s}^{-1}$ (red/blue shading) along the A–B cross section shown in Fig. 8 with (a) isentropes at 4-K intervals and (b) isotachs at 6 m s^{-1} intervals (black contours) at 1000 UTC 16 Dec 2014.

and aircraft attitude (α , q) $G_{\text{attitude}} = Z_\alpha/U_0 \alpha + (1 + Z_q/U_0)q$ (green dashed line). The G_{wg} almost follows the vertical wind changes (black solid line); however, it is apparent that G_{attitude} is generated in a direction to cancel G_{wg} . This is because the

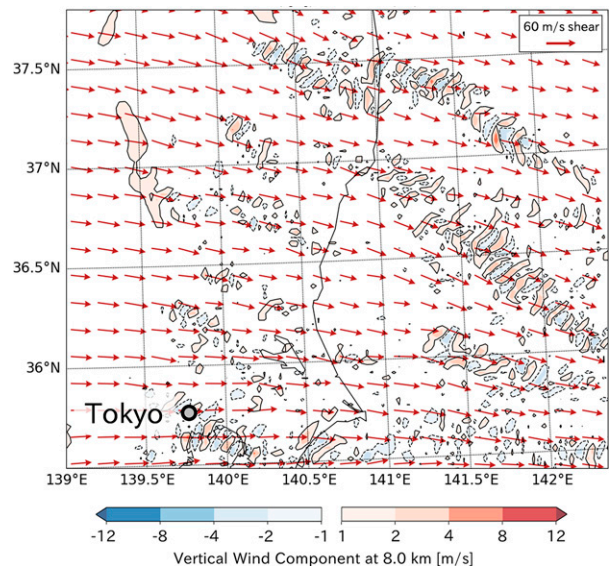


FIG. 10. Shear vectors, defined as the 6–8-km horizontal wind difference and the $|\text{vertical wind component}| > 1 \text{ m s}^{-1}$ (red/blue shading) at an altitude of 8.0 km, at 1000 UTC 16 Dec 2014 in domain 1.

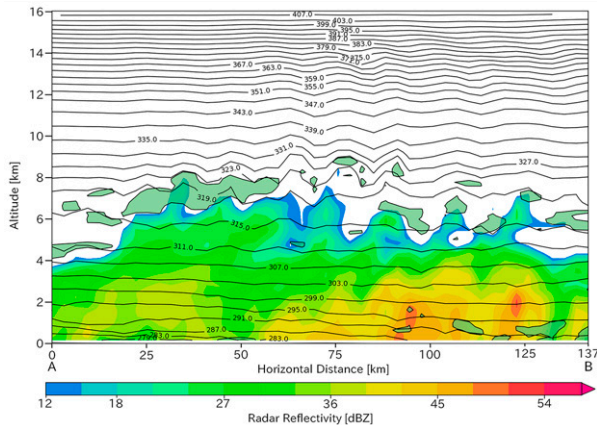


FIG. 11. WRF domain-1 simulated radar reflectivity, areas with $Ri < 0.25$ (green shading), and isentropes (K; black contours) along the A-B cross section shown in Fig. 8.

aircraft has characteristics to stabilize its attitude changes, such as α and q , in this case. In particular, Z_α is a negative value (see the appendix), which causes a strong restoring force against α changes (w change, as $\alpha \approx w/U_0$). The $G_{altitude}$ increase lags against that of G_{wg} because $G_{altitude}$ depends on α and q , which vary more slowly than w_g owing to the large aircraft mass m and moment of inertia I . Therefore, there is a slight lag between the gust and the $G_{altitude}$.

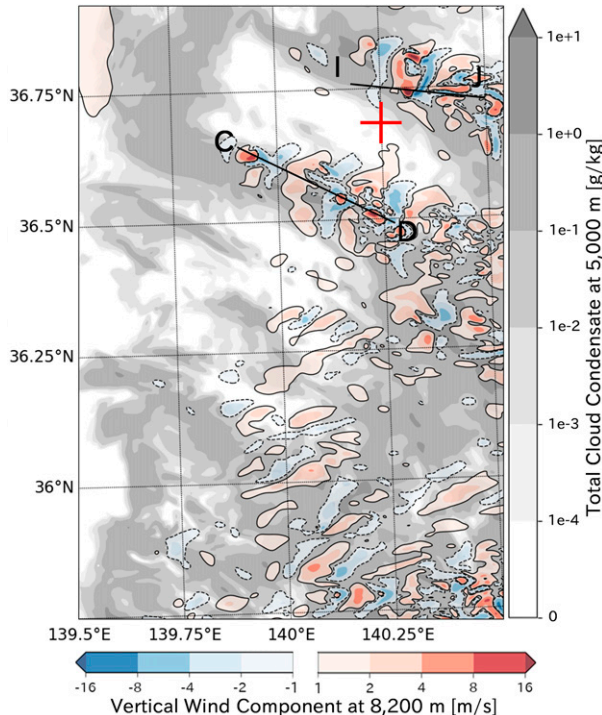


FIG. 12. [Vertical wind component] $> 1 \text{ m s}^{-1}$ (red/blue shading) at 8.2-km altitude and 5-km total cloud condensate at 1021 UTC 16 Dec 2014 simulated in domain 3. The red cross indicates the location of the target case.

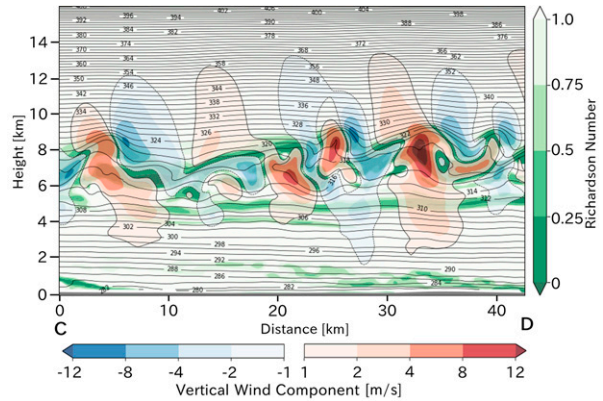


FIG. 13. Vertical cross section along C-D (Fig. 12) showing domain-3 simulated Ri (green shading), isentropes (K; black contours), and the [vertical wind component] $> 1 \text{ m s}^{-1}$ (red/blue shading).

In this study, we investigated the aircraft response to the turbulent eddies larger than that of an aircraft. By using the above method, we expect that it will become possible to check how the attitude control system introduced in section 1 behaves in response to a realistic wind field with large spatial scales.

Since severe turbulence (from $+1.8$ to -0.88 G ; Japan Transport Safety Board 2016) was reported in the target case, the aircraft should have experienced more intense shaking than was simulated in this study. To further investigate the feasibility of virtual flight testing, we would need to simulate turbulence with a grid resolution on the order of 10 m using WRF-LES.

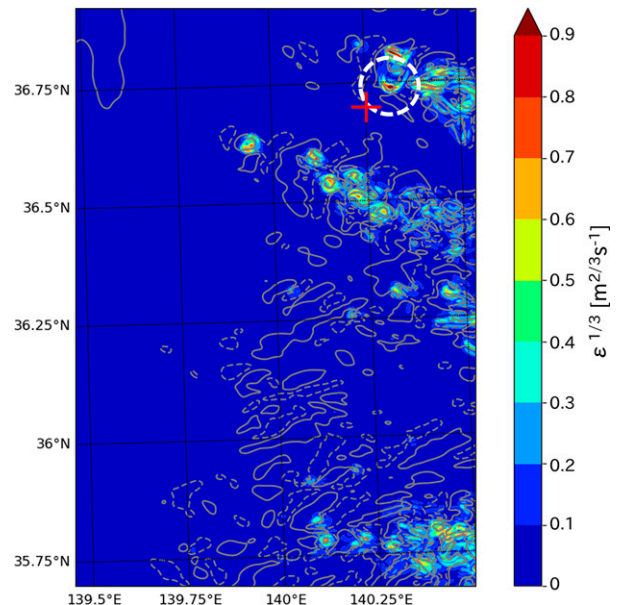


FIG. 14. EDR (ϵ) and vertical velocity $= \pm 1 \text{ m s}^{-1}$ (gray contours, with solid line being $+1 \text{ m s}^{-1}$ and dashed line being -1 m s^{-1}) at 1021 UTC 16 Dec 2014 in domain 3. Maximum EDR is found in the thick white dashed circle.

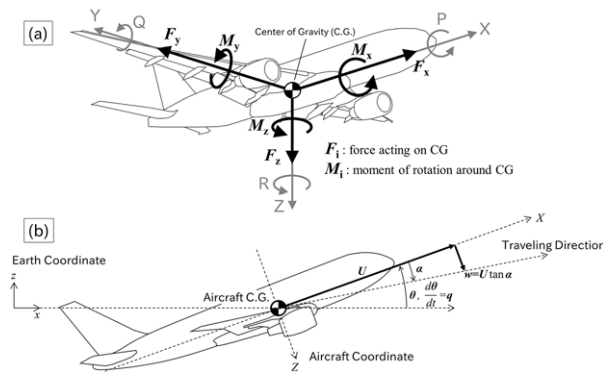


FIG. 15. (a) Forces and moments defined on the aircraft coordinate plane, and (b) state variables for aircraft motion.

6. Conclusions

In this study, we investigated the feasibility of virtual flight testing by performing a flight simulation in an airspace with turbulence reproduced using the WRF-LES model. In actual flight testing, it is not always possible to encounter intense CAT, which makes the validation of lidar and surrounding devices using a large amount of observation data difficult. If virtual flight testing in a numerically generated turbulence field is enabled, it becomes possible to conduct many flight tests under near realistic conditions, which can provide preliminary test data for future flight testing that investigates the validity of algorithms for aircraft control and forward wind estimation. However, there are almost no examples of LES for CAT at high altitudes and flight

simulations in wind fields generated by numerical weather prediction.

The target case was an accident that occurred at an altitude of 8.2 km near Tokyo on 16 December 2014. Observation data suggested that local CAT was generated in the airspace between the jet axis and precipitation clouds. First, we performed a WRF simulation to reproduce the turbulence at the time of the accident using a three-domain model configuration with the minimum horizontal grid spacing of 250 m. The synoptic-scale simulation reproduced the observed conditions well and estimated relatively small-scale (up to 15 km) waves of vertical wind speed fluctuations near the location of the accident. Furthermore, these waves were found to be instability waves because the Ri was reduced by both the vertical wind shear below the jet axis and the reduction of stability around the top of the precipitation clouds. In the LES domain, we confirmed stronger wind variations relative to the synoptic-scale simulation and found overturning isentropes in the resolved waves. The maximum EDR was over 0.7, which was classified as severe by ICAO. This indicates that the model could reproduce conditions that can cause significant aircraft shaking.

We estimated the vertical acceleration experienced by a Boeing 777-class aircraft without control by solving the aircraft equations of motion in the wind field generated by the WRF-LES. The 70-s flight simulation, which traversed the area of the maximum EDR, estimated 0.08–1.57-G acceleration changes at the point where a 32 m s^{-1} wind change was observed. This change was equivalent to moderate conditions on the AMDAR scale. Based on the promising results from this study, we expect further progress in terms of the

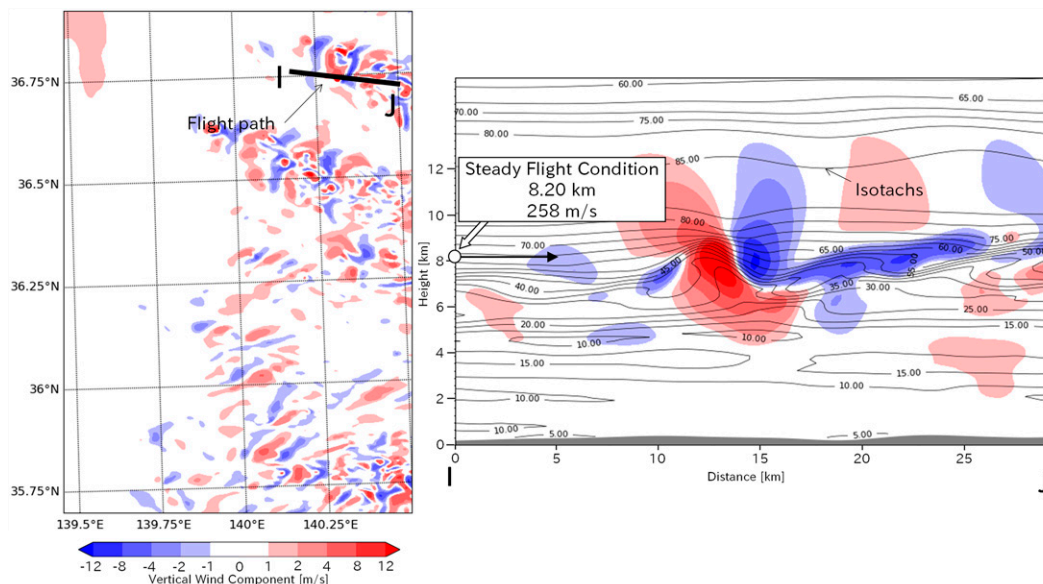


FIG. 16. Computational settings for the flight simulation. (left) The location of the cross section I-J and domain-3 simulated vertical wind component $> 1 \text{ m s}^{-1}$ (red/blue shading) at 8.2 km at 1021 UTC 16 Dec 2014. (right) the vertical cross section along I-J showing the initial location of the aircraft, the vertical wind component $> 1 \text{ m s}^{-1}$ (red/blue shading), and the horizontal wind (solid lines).

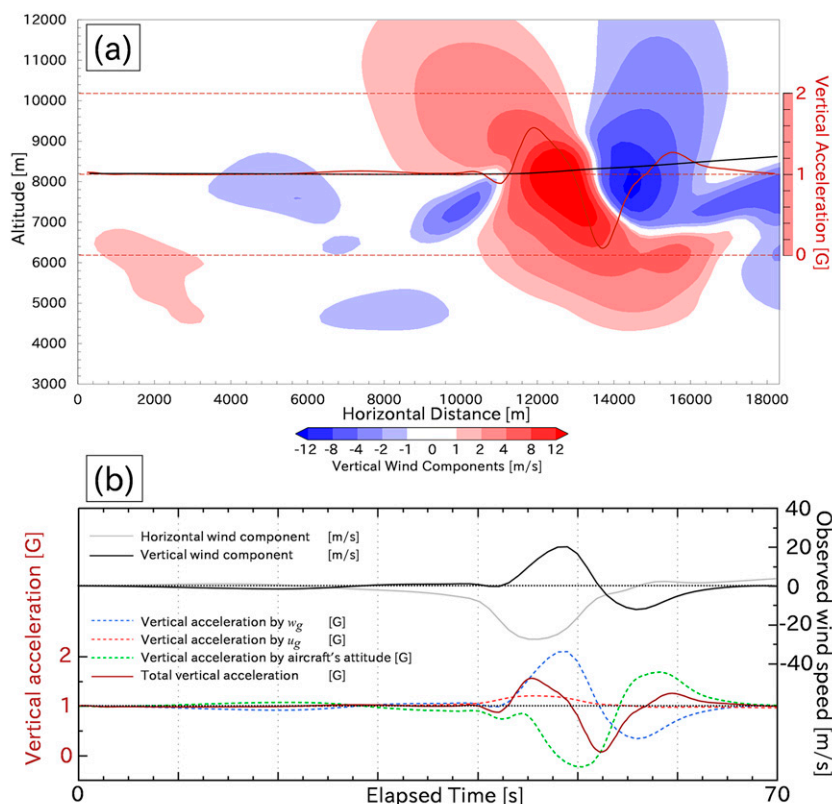


FIG. 17. (a) Simulated flight path and the time series of vertical acceleration. The vertical wind field at 1021 UTC 16 Dec 2014 is shown by red/blue shading, the flight path is a black solid line, and the history of vertical acceleration is a brown solid line. Vertical acceleration is nondimensionalized with the acceleration of gravity and has the unit of G. (b) Simulated vertical acceleration and the vertical wind components experienced by the aircraft.

feasibility of virtual flight testing based on WRF-LES and flight simulation.

In future studies, we will perform LES on a finer grid that can resolve eddies smaller than aircraft to simulate flights more accurately, for example, a turbulence that shakes a B747 at its eigenfrequency (1.29 Hz; Table A5, below) as the pitching motion shakes the aircraft with a larger amplitude than other turbulences with different frequencies. The corresponding wavelength to the eigenmotion is 200 m [$=258 \text{ (m s}^{-1})/1.29 \text{ (Hz)}$], which is smaller than the grid spacing (250 m) for WRF domain 3, suggesting that 10–20-m grid spacings may be required.

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from Mr. H. Takami of Tohoku University are greatly acknowledged.

Data availability statement. Because of confidentiality agreements, JAL PIREP data cannot be made openly available. WRF netCDF data that support the findings of this study are available from the corresponding author, Ryoichi Yoshimura, upon reasonable request.

APPENDIX

Estimating the Aerodynamic Characteristics of an Aircraft

This section describes the method used to estimate the aircraft-dependent constant coefficients included on the right-hand side of Eq. (13). We decompose the forces/moments A' to the steady part A and the unsteady part A' and assume that a linear relationship can precisely approximate these unsteady forces/moments (Z and M) within very small changes to the state variables (u , α , and q). Therefore, Eq. (13) only solves the additional motion of the aircraft from its steady flight condition. For the

example of moment M , the relationship between the additional moments and state variable disturbances is approximated linearly by Eq. (A1). These coefficients (M_u , M_α , and M_q) provide a first-order approximation for the moments/forces by multiplying with the state variable disturbances (u , α , and q):

$$\begin{aligned} M(u, \alpha, q) &= M' - \bar{M} \cong \frac{\partial M}{\partial u} u + \frac{\partial M}{\partial \alpha} \alpha + \frac{\partial M}{\partial q} q \\ &= M_u u + M_\alpha \alpha + M_q q. \end{aligned} \quad (\text{A1})$$

In the equations of motion, the total two force/moment (Z and M) depends on u , α , and q , which means there are six types of aerodynamic coefficients:

$$Z_u, Z_\alpha, Z_q, M_u, M_\alpha, \text{ and } M_q. \quad (\text{A2})$$

a. Measurement of aircraft geometry

The aircraft geometry required for coefficient estimation is measured on the basis of airport planning data from Boeing (B777-200: Boeing 2011a; B747-100: Boeing 2011b). Figure A1 shows an example of the shape of a B747-100. The outer mold line, which is gray in color, is the aerodynamic shape of a B747 (Boeing 2011b), and the shape enclosed by thick black solid lines is the simplified B747 shape composed of elements that mainly contribute to the overall aerodynamic performance. We use the latter shape to define aircraft geometry for coefficient estimation. In this study, we neglect the aerodynamics of the vertical stabilizer and engines because their contribution to the vertical acceleration changes due to turbulence is small.

The shape of the fuselage (body) is approximated by a combination of cones and cylinders, as shown in Fig. A1. The bottom shape is a circle whose area is as large as the cross-sectional area (Boeing 2011b). To estimate M_α , we estimate the moment in the pitching direction (about the y axis) using Munk's airship theory (Munk 1924). We assume tip/rear body shapes consisting of semiellipsoids (shown in Fig. A1 in blue).

The two-dimensional shape of the main wing is defined as a trapezoid whose base is located on the x axis, and its shape and the outer mold line are consistent on the outboard part of the wing. We identified the location of the mean aerodynamic chord (MAC), where the moment of the lifting force generated by the main wing becomes zero, based on point A, which is the intersection of two lines: the line defined by the measured wing tip chord length c_t and the measured wing root chord length c_r , and the 50% chord length line that connects the midpoints of the leading (front) and trailing (back) edges on the main wings. We define $x\%$ MAC as the $x\%$ location from the leading edge to the trailing edge of the MAC. The location of the aerodynamic center, on which the average lifting force of the main wing acts, and the center of gravity are set as 25% MAC and 35% MAC, respectively. We define the sweep angle Λ as the angle between the y axis and the 25% chord length line. The wing area S is 2 times as large as the area

of the trapezoid, and its aspect ratio (AR) is $1/S$ multiplied by the square of the span length b , 2 times as long as the length between the x axis and the wingtip.

The two-dimensional shape of the horizontal stabilizer is defined as a trapezoid whose base is located at the intersection of the 50% chord line and the outer mold line. Shape parameters such as MAC, S , and AR are measured in the same way as the main wing. The horizontal distance between the center of gravity and the aerodynamic center of the horizontal stabilizer is called the moment arm l_t .

b. Estimation of each coefficient

The following method follows Kato et al. (1982).

1) COEFFICIENTS FOR THE FORCE IN THE Z DIRECTION

The Z_u is the change in the z direction force with respect to airspeed u . Because the left-hand side of the equation of motion is divided by the aircraft mass m or the moment of inertia, they are included in these coefficients:

$$Z_u = \frac{\partial Z}{\partial u} = \frac{\rho U_0 S}{2m} (C_{z_u} - 2C_L). \quad (\text{A3})$$

We consider a nondimensionalized form of Z_u ; C_{z_u} is a derivative of Z_u nondimensionalized by the wing area S , the steady flight speed U_0 , and the density of air ρ . For moments, they are also nondimensionalized by MAC length $\bar{c}$. In Eq. (A3), $\rho U_0 S/m$ has a dimension of Z_u . The C_{z_u} is estimated as

$$C_{z_u} = \frac{\partial C_z}{\partial u} = -M_0 \frac{\partial C_L}{\partial M_0}, \quad (\text{A4})$$

where M_0 is U_0 divided by the sound speed. However, this coefficient can be neglected in this study. The C_L is the nondimensionalized form of the lifting force of the entire aircraft during steady flight and is estimated by nondimensionalizing the equilibrium of forces in the z direction; the lifting force is equal to the aircraft weight. In Eq. (A5), $\rho U_0^2 S$ transforms a force into a nondimensional value:

$$C_L = \frac{mg}{0.5\rho U_0^2 S}. \quad (\text{A5})$$

The Z_α is the change in the z -direction force with respect to the angle of attack α :

$$Z_\alpha = \frac{\partial Z}{\partial \alpha} = \frac{\rho U_0^2 S}{2m} C_{z_\alpha}. \quad (\text{A6})$$

The nondimensional coefficient C_{z_α} is given by α differentiation of the equilibrium of nondimensionalized force in the z direction:

$$C_{z_\alpha} = -C_{L_\alpha}. \quad (\text{A7})$$

The C_{L_α} is the α derivative of C_L . Assuming that only the main wings and horizontal stabilizers generate the lifting force of the entire aircraft, C_L is estimated as a linear function of α ; C_{L_α} is estimated as follows:

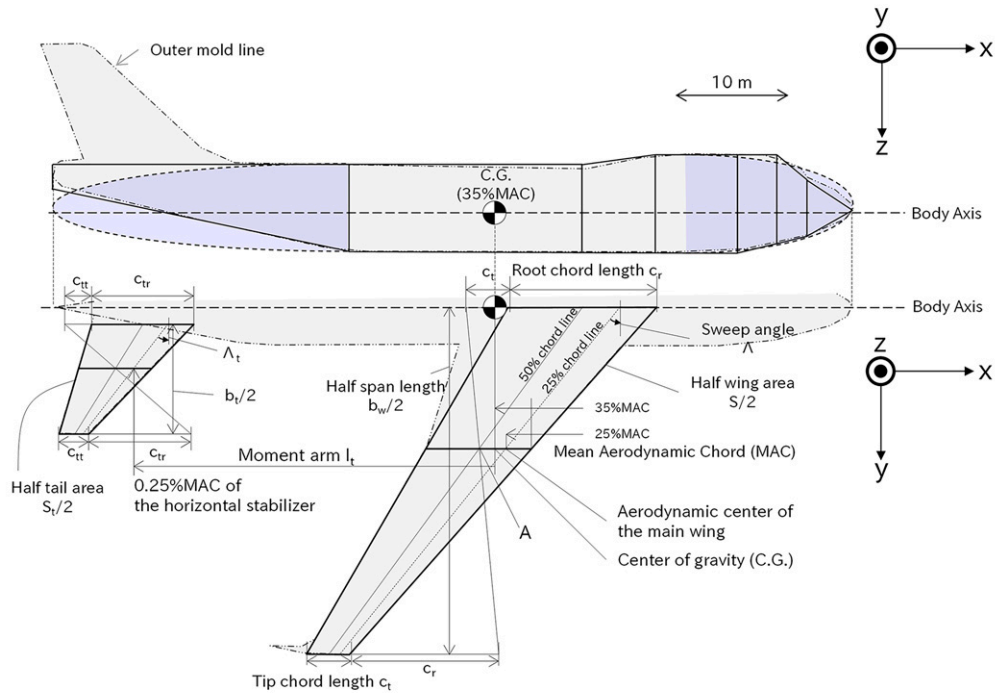


FIG. A1. Aircraft geometry.

$$C_{L_\alpha} = \frac{\partial C_L}{\partial \alpha} = a_w \left[1 + \frac{a_t}{a_w} \frac{S_t}{S} \left(1 - \frac{\partial \delta}{\partial \alpha} \right) \right]. \quad (\text{A8})$$

$$\frac{\partial \delta}{\partial \alpha} = \frac{2a_w}{\pi AR}. \quad (\text{A10})$$

The a_w and a_t are α derivatives of the nondimensionalized lifting force generated on the main wing and the horizontal stabilizer. Using the following equation, they can be empirically estimated by substituting sweep angles Λ , aspect ratios AR for each wing, and a_0 (5.7 is set in this study), which is the α derivative of the nondimensionalized lifting force of a two-dimensional airfoil:

$$a_i = \frac{a_0 \cos \Lambda_i}{1 + a_0 \frac{\cos \Lambda_i}{\pi AR_i}}. \quad (\text{A9})$$

The $\partial \delta / \partial \alpha$ is the change in downwash δ , which is the change in the direction of airflow at the location of the horizontal stabilizer brought by the aerodynamic action of the main wing, with respect to α , and is approximated by the following equation:

The Z_q is the change in the z -direction force with respect to the angular velocity around the y -axis q . The q derivative of lifting force changes due to the pitching angular velocity q gives Z_q and its nondimensional form C_{z_q} as follows:

$$Z_q = \frac{\partial Z}{\partial q} = \frac{\rho U_0 S \bar{c}}{4m} C_{z_q}, \quad C_{z_q} = -2V_h^* a_t. \quad (\text{A11})$$

The V_h^* is called the nondimensional tail volume, defined as $V_h^* = l_t S_t / (\bar{c} S)$, where l_t is the moment arm and S_t is the area of the horizontal stabilizer.

2) COEFFICIENTS FOR THE MOMENT ABOUT THE Y AXIS

The M_u is the change of the moment about the y axis with respect to the airspeed u :

TABLE A1. Estimated aerodynamics of Boeing 777-200 aircraft.

Nondimensional coefs		Dimensional coefs	
C_{z_u}	0	Z_u	-0.0757 s^{-1}
C_{m_u}	0	M_u	$0 \text{ m}^{-1} \text{ s}^{-1}$
C_{z_α}	-4.59	Z_α	-138 m s^{-2}
C_{m_α}	-0.890	M_α	-1.77 s^{-2}
$C_{m_{\dot{\alpha}}}$	-4.55	$M_{\dot{\alpha}}$	-0.148 s^{-1}
C_{z_q}	-4.38	Z_q	-2.15 m s^{-1}
C_{m_q}	-14.4	M_q	-0.468 s^{-1}

TABLE A2. Reference coefficients of Boeing 747-100 aircraft.

Nondimensional coefs		Dimensional coefs	
C_{z_u}	-0.0840	Z_u	-0.0900 s^{-1}
C_{m_u}	-0.0928	M_u	$-0.0007 \text{ m}^{-1} \text{ s}^{-1}$
C_{z_α}	-4.24	Z_α	-157 m s^{-2}
C_{m_α}	-0.629	M_α	-1.24 s^{-2}
$C_{m_{\dot{\alpha}}}$	-5.40	$M_{\dot{\alpha}}$	-0.176 s^{-1}
C_{z_q}	-5.01	Z_q	-3.05 m s^{-1}
C_{m_q}	-20.5	M_q	-0.667 s^{-1}

TABLE A3. Reference flight conditions.

Altitude, air density	6096 m, $\rho = 0.653 \text{ kg m}^{-3}$
Flight speed	Mach 0.8, $U_0 = 253 \text{ m s}^{-1}$
Lifting coef	$C_L = 0.266$
Center of gravity position	$h = 25\% \text{ MAC}$
Aircraft mass	$m = 2.89 \times 10^5 \text{ kg}$
Inertia moment about y axis	$I_{yy} = 4.49 \times 10^7 \text{ kg m}^2$
Gravity acceleration	$g = 9.80 \text{ m s}^{-2}$

$$M_u = \frac{\partial M}{\partial u} = \frac{\rho U_0 S \bar{c}}{2 I_{yy}} C_{m_u}, \quad C_{m_u} = -M_0 \frac{\partial C_M}{\partial M_0}. \quad (\text{A12})$$

We assume that this coefficient can be neglected. The M_u contributes to long-period aircraft motion. For the gust response, which is a short-period motion, the contribution of this coefficient is considered to be small.

The M_α is the change in the moment about the y axis with respect to the angle of attack α . The C_{m_α} is estimated by the α derivative of the sum of the moments around the center of gravity. We consider that the fuselage, main wing, and horizontal stabilizer generate moments:

$$M_\alpha = \frac{\partial M}{\partial \alpha} = \frac{\rho U_0^2 S \bar{c}}{2 I_{yy}} C_{m_\alpha},$$

$$C_{m_\alpha} = \frac{\partial C_M}{\partial \alpha} = a_w \left[(h - h_{nw}) - V_h^* \frac{a_t}{a_w} \left(1 - \frac{\partial \delta}{\partial \alpha} \right) \right] + M_{\text{fus}}. \quad (\text{A13})$$

Here, h and h_{nw} are the locations of the center of gravity and the aerodynamic center (% MAC) of the main wing. The term M_{fus} represents the pitching instability induced by the fuselage. We estimate this term using Munk's airship theory (Munk 1924).

The $M_{\dot{\alpha}}$ is the change in the moment about the y axis with respect to the temporal derivative of the angle of attack $\dot{\alpha} = d\alpha/dt$. The $C_{m_{\dot{\alpha}}}$ is approximated using the $\dot{\alpha}$ derivative of the increment of the downwash δ at the location of the horizontal stabilizer due to α changes as follows:

$$M_{\dot{\alpha}} = \frac{\partial M}{\partial \dot{\alpha}} = \frac{\rho U_0 S \bar{c}^2}{4 I_{yy}} C_{m_{\dot{\alpha}}}, \quad C_{m_{\dot{\alpha}}} = -2 V_h^* \left(\frac{l_t}{\bar{c}} \right) a_t \frac{\partial \delta}{\partial \alpha}. \quad (\text{A14})$$

The M_q is the change in the moment about the y axis with respect to the angular velocity around the y-axis q . Similar to the C_{z_q} estimation, C_{m_q} is obtained by the q derivative of the

TABLE A4. Estimated aerodynamics of Boeing 747-100 aircraft.

Nondimensional coefs		Dimensional coefs	
C_{z_u}	-0.00	Z_u	-0.0811 s^{-1}
C_{m_u}	-0.00	M_u	$0.00 \text{ m}^{-1} \text{ s}^{-1}$
C_{z_α}	-4.15	Z_α	-152 m s^{-2}
C_{m_α}	-0.680	M_α	-1.45 s^{-2}
$C_{m_{\dot{\alpha}}}$	-5.25	$M_{\dot{\alpha}}$	-0.201 s^{-1}
C_{z_q}	-4.62	Z_q	-3.03 m s^{-1}
C_{m_q}	-16.1	M_q	-0.617 s^{-1}

TABLE A5. Comparison of eigenmodes from the reference and estimated aerodynamics of B747-100 aircraft.

	Frequency (s^{-1})	Damping rate
Reference (Heffley and Jewell 1972)	1.29	0.57
Estimated	1.35	0.53

moment of the lifting force acting on the horizontal stabilizer generated by the angular velocity q :

$$M_q = \frac{\partial M}{\partial q} = \frac{\rho U_0 S \bar{c}^2}{4 I_{yy}} C_{m_q}, \quad C_{m_q} = -2 V_h^* \left(\frac{l_t}{\bar{c}} \right) a_t. \quad (\text{A15})$$

c. Obtained coefficients and validity of the method

Table A1 lists the estimated coefficients of forces/moments substituted into the equations of motion. We estimated coefficients for the B777-200 that encountered turbulence in the target case by using the method introduced in appendix section b above. The geometry of the B777-200 was taken from airport planning data (Boeing 2011a). The aircraft was assumed to be in a steady flight condition at 8200 m ($\rho = 0.514 \text{ kg m}^{-3}$) and $U_0 = 258 \text{ m s}^{-1}$ (Mach number $M_0 = 0.84$). The lifting coefficient was set at $C_L = 0.324$ in reference to Eq. (A5). The aircraft mass was $m = 2.47 \times 10^5 \text{ kg}$, and the moment of inertia around the y axis was estimated to be $I_{yy} = 3.15 \times 10^7 \text{ kg m}^2$ as based on the moment of inertia of the B747-100 multiplied by the mass and length ratios of the B777 and B747.

To demonstrate the validity of the method presented in section b, we investigate the estimation error between the B747-100 coefficients from Heffley and Jewell (1972) and coefficients estimated using this method. The reference coefficients of the B747-100 and its flight conditions are listed in Tables A2 and A3. The air density ρ was set from the International Standard Atmosphere.

Table A4 lists the coefficients estimated on the basis of the geometry defined using airport planning data and the method in appendix section b under the flight conditions listed in Table A3.

We compared both sets of coefficients, focusing on one of the eigenmotions of aircraft characterized by these coefficients, which is short-period, vertical motion. Because we simulated the aircraft response to gusts in this study, it was appropriate to evaluate the method and coefficients for this short-period motion. This motion is an oscillation with strong decay, where the angle of attack and the pitching angle interact with each other, and its frequency and damping rate are unique to each aircraft. Equation (13) is the equation of motion without the equation for the horizontal speed u , and the eigenmode of the equation without external force due to turbulence is only the short-period mode. The Laplace transform of the equation yields the following quadratic equation:

$$s^2 + \left(-\frac{Z_\alpha}{U_0} - M_q - M_{\dot{\alpha}} \right) s + \left(-M_\alpha + \frac{Z_\alpha}{U_0} M_q \right) = s^2 + 2\omega \xi s + \omega^2 = 0, \quad (\text{A16})$$

and we can compute the characteristic frequency ω and damping rate ξ as follows:

$$\text{frequency: } \omega = \sqrt{-M_\alpha + \frac{Z_\alpha}{U_0} M_q} \quad \text{and} \quad (\text{A17})$$

$$\text{damping rate: } \xi = \left(-\frac{Z_\alpha}{U_0} - M_q - M_\alpha \right) / (2\omega). \quad (\text{A18})$$

Table A5 compares the frequencies and damping rates obtained using our estimation and Heffley and Jewell (1972) data. The estimation errors for the frequency and damping rate are both approximately 0.05 s^{-1} , which means that the short-period motion by the coefficients obtained using our method approximates the motion well in comparison with the reference coefficients provided by Heffley and Jewell (1972). Therefore, the method presented in appendix section b is reasonable.

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